

CN Capstone Project

Mac 3 — Backend A Documentation

Phase 1 / Midsem Evidence and Implementation Record

1. Documentation Requirements

The supplied project plan identifies four main submission components: an architecture document, a configuration bundle, backend source, and an evidence folder. The evidence is expected to cover DNS, HTTPS/TLS, reverse proxy/load balancing, caching, packet analysis, and failure demonstrations. For Mac 3 specifically, the required role is Backend A on HTTP port 3001.

- Architecture document: topology, IP/service table, and request flow.
- Configuration bundle: relevant DNS/nginx configuration plus backend source.
- Backend source: Backend A implementation.
- Evidence folder: numbered screenshots and the Wireshark capture for the group demonstration.
- Mac 3 evidence: listening service, root endpoint, status endpoint, and the later Backend-A failure demonstration.

2. Project Topology

Machine	IP	Role	Service
Mac 1	10.7.9.245	DNS + client	dnsmasq :53
Mac 2	10.7.5.53	Edge / reverse proxy / LB	nginx HTTPS :8443
Mac 3	10.7.22.10	Backend A	HTTP :3001
Mac 4	10.7.7.25	Backend B + client	HTTP :3002

The intended request path is: Mac 4 client → Mac 1 DNS → Mac 2 nginx HTTPS edge → Backend A (Mac 3) or Backend B (Mac 4).

3. Mac 3 Role and Service

- Hostname observed: Rachits-MacBook-Pro-4.local
- Network interface: en0
- Mac 3 IP address: 10.7.22.10
- Default gateway observed: 10.7.0.1
- Backend service: Backend A
- Transport/service port: TCP 3001
- Binding: 0.0.0.0:3001, allowing LAN access rather than localhost-only access.

4. Backend A Implementation

Backend A is a lightweight Python HTTP service. It provides the required root and status endpoints and exposes response headers used by the project to identify the backend and demonstrate HTTP caching.

Key configuration:

```
HOST = "0.0.0.0"  
PORT = 3001  
BACKEND = "A"  
ETAG = '"A-v1"'
```

Required endpoints:

```
GET /  
GET /api/status
```

Important response headers:

```
X-Backend: A  
Cache-Control: max-age=60  
ETag: "A-v1"
```

5. Verification Procedure

The following commands were used to demonstrate the Mac 3 service:

```
ipconfig getifaddr en0  
  
lsof -nP -iTCP:3001 -sTCP:LISTEN  
  
curl -i http://127.0.0.1:3001/  
  
curl -i http://127.0.0.1:3001/api/status  
  
curl -i -H 'If-None-Match: "A-v1"' http://127.0.0.1:3001/api/status
```

6. Verification Results

- IP discovery returned 10.7.22.10 on en0.
- Lsof showed Python listening on TCP *:3001.
- GET / returned HTTP 200 OK and X-Backend: A.
- GET /api/status returned HTTP 200 OK and X-Backend: A.
- The status response included Cache-Control: max-age=60 and ETag: "A-v1".
- A conditional request using If-None-Match: "A-v1" returned HTTP 304 Not Modified.
- Mac 2 successfully reached http://10.7.22.10:3001/api/status and received X-Backend: A.
- A direct DNS query to Mac 1 at 10.7.9.245 resolved app.cn-capstone.test to 10.7.5.53.

7. Caching Demonstration

The caching requirement is demonstrated using an ETag and a conditional request. The first status request returns the resource together with ETag: "A-v1". A later request supplies that ETag through If-None-Match. When the representation has not changed, Backend A returns 304 Not Modified rather than returning the response body again.

```
curl -i http://127.0.0.1:3001/api/status  
  
curl -i -H 'If-None-Match: "A-v1"' http://127.0.0.1:3001/api/status
```

8. LAN Reachability / Edge Integration

Mac 2 (the edge machine) successfully accessed Backend A using Mac 3's LAN address:

```
curl -i http://10.7.22.10:3001/api/status
```

The observed response was HTTP 200 OK with X-Backend: A and the Backend A JSON status. This proves that Backend A is not limited to localhost and can be reached from another machine on the private network.

9. DNS Observation

Mac 3 was unable to resolve app.cn-capstone.test using its default resolver configuration at the time of testing. However, querying Mac 1's DNS server directly succeeded:

```
dig @10.7.9.245 app.cn-capstone.test
```

The direct query returned 10.7.5.53. The attempt to change Mac 3's system DNS using sudo was not completed because the account did not have sudoers/admin permission. No unsupported workaround was used.

10. Evidence Checklist for Mac 3

Evidence file	What it proves	Status
14-Mac3-Backend-A-Listening.png	lsof showing TCP *:3001 (LISTEN)	Completed
15-Mac3-Backend-A-Root.png	GET / returns HTTP 200 and X-Backend: A	Completed
16-Mac3-Backend-A-Status.png	GET /api/status returns HTTP 200, X-Backend: A, caching headers	Completed
17-Mac3-Backend-A-Stopped.png	Backend A stopped for the later failure/failover demonstration	Pending / only do after normal evidence

11. Live Video Demonstration Script

Recommended narration for the Mac 3 section:

"This is Mac 3, which is Backend A. Its IP address is 10.7.22.10. Backend A is listening on TCP port 3001. I am now sending a request to the status endpoint, and the response shows X-Backend: A, confirming that the request reached Backend A."

Commands to show on screen:

```
ipconfig getifaddr en0
lsof -nP -iTCP:3001 -sTCP:LISTEN
curl -i http://127.0.0.1:3001/api/status
```

12. Failure Demonstration — Do Later

The project plan reserves the Backend A stop/failure test for after all normal Phase 1 evidence is complete. At that point, Backend A is stopped with Ctrl+C while the team tests the edge. Successful requests should continue through Backend B. Backend A should then be restarted using the normal launch command.

- Do not stop Backend A during the normal Mac 3 recording.
- Do not change nginx during the failure demonstration.
- Capture the stopped state only when the team is ready to demonstrate failover.

13. Final Mac 3 Status

- Backend A is operational on 10.7.22.10:3001.
- The service is LAN-accessible.
- GET / works.
- GET /api/status works.
- X-Backend: A is present.
- Cache-Control and ETag are present.
- Conditional caching returns 304 Not Modified.
- Mac 2 can reach Backend A.
- DNS server Mac 1 is reachable directly from Mac 3.
- Backend-A failure/failover evidence remains to be captured during the group demonstration.

14. Group Submission Checklist

- Architecture document: topology, IP/service table, and request flow.
- Configuration bundle: dnsmasq configuration, nginx configuration, and backend source.
- Evidence folder: Mac 1, Mac 2, Mac 3, Mac 4, and Wireshark screenshots.
- Wireshark capture: cn-capstone-phase1.pcapng.
- Live demonstrations: DNS resolution, HTTPS, A/B load balancing, caching/304, packet analysis, and Backend A failure/failover.

Prepared from the supplied CN Capstone Phase 1 execution/evidence plan and the recorded Mac 3 test results.